Remainders and last digits

L2-NT2 Level 2 Intermediate · Number theory

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How to compute remainders and last digits of huge numbers using remainder arithmetic, cycles of powers and digit tests.
By the end you can find remainders of large sums, products and powers without computing them; use cycles of last digits to find the last one or two digits of a power; use digit-sum, alternating-sum and last-digits tests to handle unknown digits.
Prerequisites: Factors, multiples and primes (L1-NT1), Factors, primes and divisibility (L2-NT1).
Contests: IMC, Grey Kangaroo, Pink Kangaroo, Cayley, Hamilton, JMO, AMC 8, AMC 10. **Time:** about 75 minutes.

1 Lesson

What is the last digit of 3^{2026}? The number has 967 digits, so instead we look at the last digits of the powers of 3. The numbers $3, 9, 27, 81, 243, 729, \ldots$ end in $3, 9, 7, 1, 3, 9, \ldots$. The last digit of the next power depends only on the last digit of the current one (multiply by 3 and keep the last digit), so once we are back at 3 the pattern must repeat, with period 4. Since $2026 = 4 \times 506 + 2$, the power 3^{2026} sits at the same point of the cycle as 3^2, and its last digit is 9.

The argument used two facts, and this unit develops both. The remainder of a product depends only on the remainders of the factors, and since there are only finitely many remainders, powers must eventually cycle.

1.1 Remainder arithmetic

Dividing an integer a by a positive integer n gives $a = qn + r$ with $0 \le r < n$, and r is the remainder. When a and b leave the same remainder on division by n we write $a \equiv b \pmod{n}$, read "a is congruent to b mod n". This is shorthand for "n divides $a - b$". For example $17 \equiv 2 \pmod{5}$ and $100 \equiv -1 \pmod{101}$.

Remainders respect addition and multiplication.
If $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then

$$a + b \equiv a' + b', \qquad a - b \equiv a' - b', \qquad ab \equiv a'b' \pmod{n},$$

and therefore $a^k \equiv (a')^k \pmod{n}$ for every positive integer k.

Proof. Write $a = a' + sn$ and $b = b' + tn$. Then $a + b = (a' + b') + (s + t)n$ and $ab = a'b' + (a't + b's + stn)n$, so in each case the difference is a multiple of n. The statement about powers follows by multiplying $a \equiv a'$ by itself k times.

So in any calculation built from $+$, $-$ and $\times$, you may replace a number by anything with the same remainder, at any stage. The remainder of 1234×5678 on division by 10 is the remainder of $4 \times 8 = 32$, which is 2. Two consequences are used constantly.

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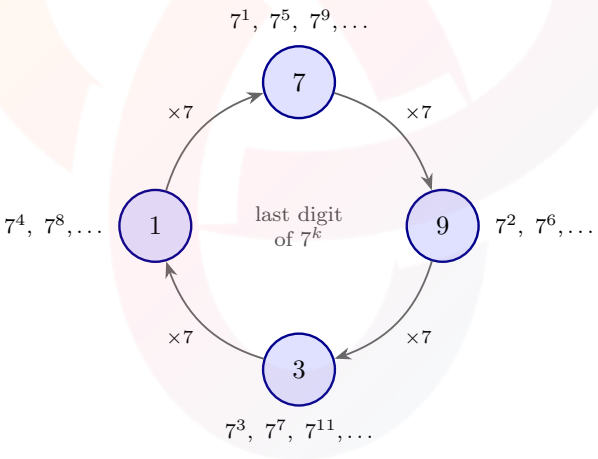
- You may reduce as often as you like, so the numbers never get big.
- You may use negative representatives. Mod 9, the number 8 is -1, so $8^{101} \equiv (-1)^{101} = -1 \equiv 8 \pmod 9$. Mod 13, $5^2 = 25 \equiv -1$. Most questions in this unit become short once you find a small power that is ± 1.

What you may *not* do is reduce the exponent mod n. Mod 5 we have $7 \equiv 2$, but $2^7 = 128 \equiv 3$ while $2^2 = 4$. Exponents can be reduced only by the length of a cycle, which is usually a different number.

1.2 Cycles of last digits

The last digit of a number is its remainder mod 10, so last-digit questions are remainder questions mod 10. The cycles for powers are worth remembering.

- A last digit of 0, 1, 5 or 6 means every power ends in the same digit.
- A last digit of 4 gives powers ending $4, 6, 4, 6, \ldots$, and a last digit of 9 gives powers ending $9, 1, 9, 1, \ldots$. Both have period 2.
- Last digit 2 gives $2, 4, 8, 6$. Last digit 3 gives $3, 9, 7, 1$. Last digit 7 gives $7, 9, 3, 1$. Last digit 8 gives $8, 4, 2, 6$. All have period 4.



Why powers cycle.
Fix a and n. The remainder of a^{k+1} mod n is determined by the remainder of a^k (multiply by a and reduce). There are only n possible remainders, so two of $a^1, a^2, \ldots, a^{n+1}$ leave the same remainder, and from then on the sequence repeats. For last digits every period is 1, 2 or 4, so **for $k \geq 1$, the last digit of a^k depends only on the last digit of a and the remainder of k mod 4**, where a remainder of 0 means the fourth term of the cycle.

Last two digits work in the same way, mod 100. The cycles are longer, so look for a power that ends in 01 or 99. For instance $7^2 = 49$ and $7^4 = 2401$, so the last two digits of the powers of 7 run $07, 49, 43, 01$ and then repeat.

1.3 Digit-sum tests and their relatives

A number with digits $\ldots d_2 d_1 d_0$ has value $d_0 + 10d_1 + 100d_2 + \cdots$. Every divisibility test comes from asking what 10 and its powers are mod the divisor.

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Tests from place value.

- Since $10 \equiv 1 \pmod 9$, every power of 10 is $\equiv 1$, so **a number is congruent to its digit sum mod 9**, and therefore also mod 3. This gives the remainder itself, not just a yes or no.
- Since $10 \equiv -1 \pmod{11}$, a number is congruent mod 11 to its alternating digit sum $d_0 - d_1 + d_2 - \cdots$, starting from the units digit.
- Since 10 is a multiple of 2 and 5, $100 = 4 \times 25$ and $1000 = 8 \times 125$, the remainder mod 2 or 5 depends only on the last digit, mod 4 or 25 on the last two digits, and mod 8 or 125 on the last three.
- Since $1001 = 7 \times 11 \times 13$, we have $1000 \equiv -1$ mod 7, 11 and 13, so split the number into blocks of three digits from the right and alternate the signs.

For instance $2026 \equiv 2 + 0 + 2 + 6 = 10 \equiv 1 \pmod 9$, and $2026 \equiv 6 - 2 + 0 - 2 = 2 \pmod{11}$. To test for a composite number, split the number into pieces with no common factor, as in $72 = 8 \times 9$ and $99 = 9 \times 11$. A number divisible by two such pieces is divisible by their product, which is the LCM fact from L2-NT1.

Example 1.
What is the last digit of $7^{2026} + 3^{2026}$?

Solution.
We take each term with its own cycle. For 7 the cycle is $7, 9, 3, 1$, and $2026 \equiv 2 \pmod 4$, so 7^{2026} ends in 9. For 3 the cycle is $3, 9, 7, 1$, and position 2 again gives 9. So the sum ends in the last digit of $9 + 9 = 18$, which is 8.
Alternatively, note that since $7 \equiv -3 \pmod{10}$, we get $7^{2026} \equiv (-3)^{2026} = 3^{2026}$, because the exponent is even. So the two terms always share a last digit here, and only one cycle was needed.

Example 2.
The five-digit number $\overline{4A27B}$, with digits $4, A, 2, 7, B$ in that order, is divisible by 99. Find the number.

Solution.
There is no test for 99 as such, but $99 = 9 \times 11$ and 9 and 11 have no common factor. So we need divisibility by 9 and by 11, and each of those has a test.
For 9, the digit sum $4 + A + 2 + 7 + B = 13 + A + B$ must be a multiple of 9. Since $0 \le A + B \le 18$, this means $A + B = 5$ or $A + B = 14$.
For 11, the alternating sum from the units digit is $B - 7 + 2 - A + 4 = B - A - 1$, and this must be a multiple of 11. But $B - A - 1$ lies between -10 and 8, so the value must be 0, which gives $B = A + 1$.
Then $A + B = 2A + 1$ is odd, which rules out 14. So $2A + 1 = 5$, giving $A = 2$ and $B = 3$, and the number is $42273 = 99 \times 427$. The step to remember is the bounding (digits are between 0 and 9). A condition mod 9 or mod 11 on digits usually leaves only one or two real possibilities, because the digit sums are small.

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Example 3.
Find the remainder when $5^{2026} + 8^{2026}$ is divided by 13.

Solution.
Listing powers of 5 mod 13 until we return to 1 would work, but first we check small powers for ± 1. We have $5^2 = 25 = 26 - 1 \equiv -1 \pmod{13}$, and also $8^2 = 64 = 65 - 1 \equiv -1 \pmod{13}$.
Now $5^{2026} = (5^2)^{1013} \equiv (-1)^{1013} = -1$, and in the same way $8^{2026} \equiv -1$. So the sum is congruent to -2 mod 13. A remainder must lie between 0 and 12, so we convert to $-2 \equiv 11 \pmod{13}$. The answer is 11.
Without this observation, we would find $5^1, 5^2, 5^3, 5^4 \equiv 5, 12, 8, 1$, a cycle of length 4, and $2026 \equiv 2 \pmod 4$ gives 12 again, so the answer is the same.

Example 4.
What are the last two digits of 3^{2026}?

Solution.
We want the remainder of 3^{2026} mod 100. The last digit has a cycle of length 4, but mod 100 the cycle is longer, and listing the cycle one term at a time is error-prone. Instead, we look for a power of 3 that is ± 1 mod 100. Repeated squaring reaches high powers quickly. Here $3^5 = 243 \equiv 43$, so $3^{10} \equiv 43^2 = 1849 \equiv 49$, and then $3^{20} \equiv 49^2 = 2401 \equiv 1 \pmod{100}$.
Now that a power is 1, exponents can be reduced by 20. As $2026 = 20 \times 101 + 6$, we get $3^{2026} = (3^{20})^{101} \times 3^6 \equiv 3^6 = 729 \equiv 29 \pmod{100}$. The last two digits are 29.
Check against the last-digit cycle. Here $2026 \equiv 2 \pmod 4$ predicts a last digit of 9, and 29 does end in 9. After a long calculation, always run a quick check like this.

Common traps.

- An exponent that is a multiple of 4 gives the *last* term of a period-4 cycle, so 2^{100} ends in 6, not 2.
- Reduce the base mod n freely, but reduce the exponent only by the length of a cycle.
- A final remainder must be between 0 and $n - 1$. If your working gives -2 mod 13, the answer is 11.
- The digit-sum test works for 3 and 9 only. The number 1899 has digit sum 27 but is not a multiple of 27.
- You cannot cancel freely, since $2 \times 3 \equiv 2 \times 8 \pmod{10}$, but $3 \not\equiv 8 \pmod{10}$. Division needs care, and Level 3 deals with division properly.
- For a difference, borrow. If a ends in 3 and b ends in 7, with $a > b$, then $a - b$ ends in $3 - 7 + 10 = 6$.

1.4 How to spot this in a contest

Any question that asks for a last digit, the last two digits, a remainder or a day of the week is remainder arithmetic. Huge powers or huge expressions, such as 2026^{2026}, $1! + 2! + \cdots + 100!$ or a long number made by writing digits in a row, are a sign that the question cannot want the number itself, only its remainder. In a factorial sum, look for the point after which every term is divisible by n. Only the first few terms then matter. Unknown digits in a number divisible by 36,

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45, 72 or 99 mean split the divisor into pieces with no common factor and apply a test to each. For repeating blocks like 202620262026, ask which power of 10 is ± 1.

When an answer seems to need a large computation, we have the following strategies:

1. reduce the base early and often,
2. use -1 wherever possible, and look for a small power that is ± 1,
3. check the result with a second modulus, such as the last digit.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ What is the units digit of $2^{2026} + 2026^2$?

Problem 2 ⋆⋆ Today is a Tuesday. What day of the week is 10^{10} days from today?

Problem 3 ⋆⋆ The four-digit number $\overline{A26B}$, with digits $A, 2, 6, B$ in that order, is divisible by 36. What is the largest value the number can take?

Problem 4 ⋆⋆⋆ What is the remainder when $2^{100} + 3^{100}$ is divided by 13?

Problem 5 ⋆⋆⋆ What is the units digit of $1^2 + 2^2 + 3^2 + \cdots + 2026^2$?

Problem 6 ⋆⋆⋆⋆ The number N_n is formed by writing the block 2026 n times in a row, so $N_1 = 2026$, $N_2 = 20262026$, and so on. What is the smallest positive integer n for which N_n is divisible by 99?

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 8 2024 Problem 1** ⋆
   Only the units digits matter.
2. **UKMT IMC 2017 Question 3** ⋆
   Which digits can end a square?
3. **AMC 8 2018 Problem 7** ⋆
   Digit sum for 9. The last three digits for 8.
4. **UKMT IMC 2021 Question 13** ⋆⋆
   Replace each factor by its digit sum.
5. **UKMT IMC 2025 Question 21** ⋆⋆
   Reduce the base first, then look for a power that is plus or minus 1.

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6. **AMC 10A 2019 Problem 2** ⋆⋆
   Count the factors of 2 and 5 in each factorial.

7. **AMC 10B 2017 Problem 14** ⋆⋆
   Group the integers by their remainder mod 5.

8. **AMC 10A 2018 Problem 19** ⋆⋆⋆
   Cycles of last digits. Reduce the exponent mod 4.

9. **UKMT IMC 2018 Question 20** ⋆⋆⋆
   Split 72 into pieces with no common factor and test each.

10. **AMC 10B 2019 Problem 14** ⋆⋆⋆
    The tests for 9 and 11 give two equations in the missing digits.

11. **UKMT Cayley 2022 Question 2** ⋆⋆⋆
    Split 45 into 5 and 9. Think about what each forces on a palindrome.

12. **AMC 10B 2017 Problem 23** ⋆⋆⋆⋆
    Find the remainder mod 5 and mod 9 separately, then combine.

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4 Hints

1. Only last digits matter. Find where 2026 sits in the cycle of last digits of the powers of 2.
2. Days repeat every 7, so you need the remainder of 10^{10} on division by 7. Replace 10 by 3 and look for a power of 3 that is ± 1 mod 7.
3. Write $36 = 4 \times 9$ and use one test for each piece. The test for 4 restricts B straight away.
4. Look for small powers of 2 and of 3 that are ± 1 mod 13.
5. The last digit of n^2 depends only on the last digit of n, so the last digits of the squares repeat every 10 terms.
6. Split 99 into 9 and 11. For 11, note that $N_n = 2026 \times (1 + 10^4 + 10^8 + \cdots)$ and ask what 10^4 is mod 11.

5 Solutions

Problem 1

What is the units digit of $2^{2026} + 2026^2$?

The powers of 2 end in $2, 4, 8, 6$, repeating with period 4. Since $2026 = 4 \times 506 + 2$, the power 2^{2026} ends in the same digit as 2^2, namely 4. The number 2026^2 ends in the last digit of $6^2 = 36$, namely 6. The sum ends in the last digit of $4 + 6 = 10$, so the units digit is 0.

Answer: 0

Problem 2

Today is a Tuesday. What day of the week is 10^{10} days from today?

We need the remainder of 10^{10} mod 7. Since $10 \equiv 3 \pmod 7$, this is the remainder of 3^{10}. Now $3^3 = 27 = 28 - 1 \equiv -1 \pmod 7$, so $3^9 = (3^3)^3 \equiv (-1)^3 = -1$, and $3^{10} \equiv -3 \equiv 4 \pmod 7$. So 10^{10} days is a whole number of weeks plus 4 days, and 4 days after a Tuesday is a Saturday.

Answer: Saturday

Problem 3

The four-digit number $\overline{A26B}$, with digits $A, 2, 6, B$ in that order, is divisible by 36. What is the largest value the number can take?

Since $36 = 4 \times 9$ and 4 and 9 have no common factor, we need divisibility by both. For 4, the last two digits $\overline{6B}$ must form a multiple of 4, so $\overline{6B}$ is 60, 64 or 68, and B is 0, 4 or 8. For 9, the digit sum $A + 8 + B$ must be a multiple of 9, where $1 \le A \le 9$. If $B = 0$ then $A + 8$ is a multiple of 9, so $A = 1$, giving 1260. If $B = 4$ then $A + 12$ is a multiple of 9, so $A = 6$, giving 6264. If $B = 8$ then $A + 16$ is a multiple of 9, so $A = 2$, giving 2268. The largest is 6264.

Answer: 6264

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Problem 4

What is the remainder when $2^{100} + 3^{100}$ is divided by 13?

We have $2^6 = 64 = 65 - 1 \equiv -1 \pmod{13}$, so $2^{12} \equiv 1$. As $100 = 12 \times 8 + 4$, we get $2^{100} = (2^{12})^8 \times 2^4 \equiv 16 \equiv 3 \pmod{13}$. Also $3^3 = 27 = 26 + 1 \equiv 1 \pmod{13}$, so $3^{99} = (3^3)^{33} \equiv 1$ and $3^{100} \equiv 3$. The sum is congruent to $3 + 3 = 6$, so the remainder is 6.

Answer: 6

Problem 5

What is the units digit of $1^2 + 2^2 + 3^2 + \cdots + 2026^2$?

The last digit of n^2 depends only on the last digit of n. For n ending in $1, 2, \ldots, 9, 0$ the squares end in $1, 4, 9, 6, 5, 6, 9, 4, 1, 0$, whose sum is 45. The numbers 1 to 2020 make up 202 complete blocks of ten, so the last digits of their squares add up to $202 \times 45 = 9090$, which ends in 0. The remaining terms $2021^2, \ldots, 2026^2$ end in $1, 4, 9, 6, 5, 6$, which add to 31. So the whole sum ends in the same digit as $0 + 31$, and the units digit is 1.

Answer: 1

Problem 6

The number N_n is formed by writing the block 2026 n times in a row, so $N_1 = 2026$, $N_2 = 20262026$, and so on. What is the smallest positive integer n for which N_n is divisible by 99?

Since $99 = 9 \times 11$ and 9 and 11 have no common factor, N_n is divisible by 99 exactly when N_n is divisible by both 9 and 11.

For 9, each block contributes digit sum $2 + 0 + 2 + 6 = 10$, so the digit sum of N_n is $10n$, and $N_n \equiv 10n \equiv n \pmod{9}$. So 9 divides N_n exactly when 9 divides n.

For 11, we have $N_n = 2026 \times (1 + 10^4 + 10^8 + \cdots + 10^{4(n-1)})$. Since $10 \equiv -1 \pmod{11}$, every $10^{4j} \equiv 1$, so the bracket is congruent to n. Also $2026 \equiv 6 - 2 + 0 - 2 = 2 \pmod{11}$. Hence $N_n \equiv 2n \pmod{11}$. As 11 is prime and does not divide 2, Euclid's lemma shows that 11 divides $2n$ exactly when 11 divides n.

So we need both 9 and 11 to divide n, that is, 99 divides n. The smallest such n is 99.

Answer: 99

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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